Experiment 10

Ex. No.: 10

Extract EXIF Data from Image Files Using ExifReader Software

Aim

To inspect the image properties and available EXIF-related metadata of a JPEG file.

Tools required

1. ExifReader on Windows for the manual workflow.
2. A JPEG image from an authorised source.
3. macOS tools used here: `sips`, `file`, and `shasum`.
4. A synthetic JPEG created for this report; it contains no personal location or camera data.

Idea/Theory

1. EXIF is metadata stored in supported image files and may describe camera make/model, capture time, exposure, shutter speed, flash, orientation, software, and GPS coordinates.
2. Not every JPEG contains every EXIF tag; exported or edited images may have only basic format metadata.
3. Metadata can be altered, removed, or inconsistent, so important values should be corroborated.
4. A hash records the exact examined byte stream and helps distinguish metadata review from later file changes.

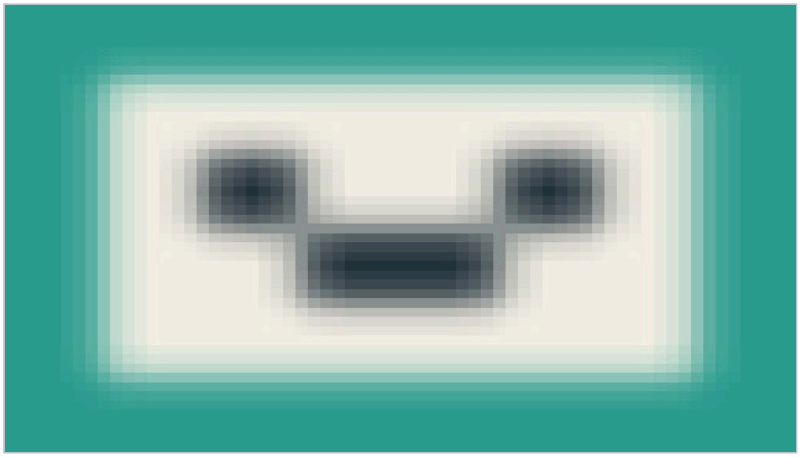
Experiment 10 — continued

Procedure/Steps

1. Install and open `ExifRead.exe` on Windows.
2. Select **Open** and browse to the authorised JPEG or JPG file.
3. Review the thumbnail and the EXIF fields displayed by ExifReader.
4. Record only relevant fields, including their exact values and the source file hash.
5. For the executed macOS equivalent, run the commands below from `evidence/exp10/`.

```
sips -g all sample.jpg
file sample.jpg
shasum -a 256 sample.jpg
```

Executed on macOS using sips, file, and shasum on sample files created for this report (the manual's Windows tool: ExifReader).



Synthetic JPEG used for the recorded output; no personal data.

Experiment 10 — continued

Observation/Output

```
sample.jpg
  pixelWidth: 640
  pixelHeight: 360
  typeIdentifier: public.jpeg
  format: jpeg
  formatOptions: default
  dpiWidth: 72.000
  dpiHeight: 72.000
  samplesPerPixel: 3
  bitsPerSample: 8
  hasAlpha: no
  space: RGB
  profile: sRGB IEC61966-2.1

File identification:
sample.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 72x72, segment length 16, Exif
Standard: [TIFF image data, big-endian, direntries=1], baseline, precision 8, 640x360, components 3

SHA-256:
ad51262a244a2a78645f587d7b6f2828a1e63f4dec82073c87a5d4cb7e769559  sample.jpg
```

Result

The real `sips` output reported a 640×360 JPEG with 72 dpi, 8-bit RGB samples, no alpha channel, and an sRGB profile; `file` also identified a JFIF JPEG containing a small EXIF-standard TIFF directory. This demonstrated extraction of the metadata actually present in the synthetic image; no camera, timestamp, or GPS value was claimed.